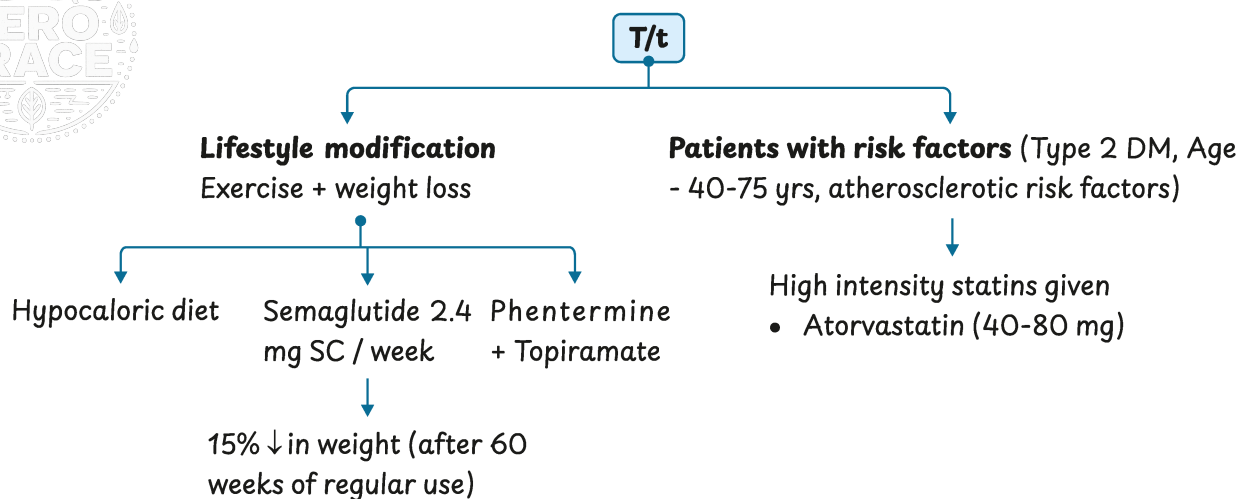


Important Information

- In patients with MI event, severe CKD (eGFR <30 ml/min)
 - Target LDL - <55 mg/dl
- In patients with multiple risk factors and moderate CKD (eGFR <60 ml/min)
 - Target LDL - <70 mg/dl

SYNDROME Z

00:14:28

- Syndrome X + Obstructive sleep apnea
 - Airway collapse → Multiple apneic episodes/hour → Sleep interruption
- Investigation - Polysomnography
 - > 15 apneic episodes / hour → Obstructive sleep apnea
 - Intervention - C-PAP

CARDIAC SYNDROME X / MICROVASCULAR ANGINA

- Pathology
 - Perforators narrowing due to atherosclerotic risk factors
 - Epicardial coronary artery → Normal
- Features
 - Chest pain on exertion/emotion - Chronic stable angina
- Investigation
 - Treadmill test
 - ST depression ≥1 mm persisting for ≥80 msec in 2 leads
 - Coronary/ Conventional CT angiography → Normal
- T/t
 - ↓ dose Aspirin + Isosorbide mononitrate + Metoprolol + Statin



Q. Which of the following is not a component of syndrome Z?

- a. HDL cholesterol <40 mg/dl in men
- b. Blood pressure $>130/85$ mm Hg
- c. Fasting triglyceride >150 mg/dl
- d. Waist/Hip ratio

Ans: (d)

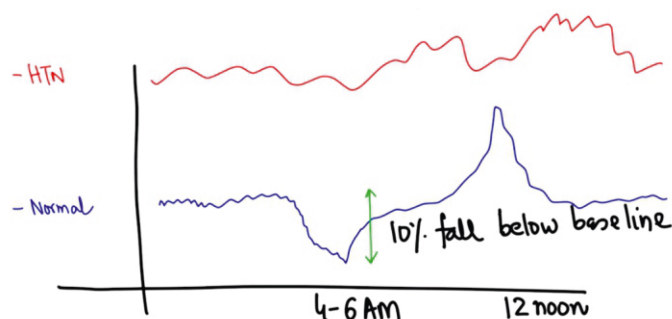


TR. HYPERTENSION



BP VARIATION IN NORMAL VS HYPERTENSIVE PATIENTS

00:01:32



- BP is lowest between 4-6 am
 - 10% of fall below baseline value
- BP is highest between 12 noon
- Vasoconstriction is maximum in morning hours → ↑ stress on the heart → ↑ chances of ACS, STEMI, NSTEMI, Unstable angina

INTRODUCTION

00:04:40

- M/c non-communicable disease: **Hypertension**
- M/c cause of death in HTN: **Cardiovascular mortality**
 - LVH → ↑ oxygen demand while supply remains normal → Demand-supply mismatch
- Incidence: 45%
- Early identification of (increasing incidence due to lifestyle changes) HTN: Perform **ABPM** → **Detects loss of nocturnal dip**
- Advantages of good BP control:
 - ↓ ACS incidence by 25%
 - ↓ Stroke incidence by 30%
 - ↓ CHF incidence by 50%

CLASSIFICATION

00:09:27

- Optimal BP: **~115/75 mmHg**

AHA GUIDELINES	
Elevated	120-129/<80 (2017) mmHg
Stage 1	130-139/80-89 mmHg
Stage 2	≥140/≥90 mmHg

BVA SPOCC CUFF

- Readings are inversely proportional to cuff size (smaller cuff → higher BP reading)
- Cuff length: >80% of the arm circumference
- Cuff width: >40% of the arm circumference
- Measurement tips:
 - Record after 5 min of rest
 - Avoid caffeine and cigarettes for >30 min before recording
 - Check BP in both left and right arms, and lower extremities



WHITE COAT HYPERTENSION VS MASKED HYPERTENSION

00:15:28

WHITE COAT HYPERTENSION	MASKED HYPERTENSION
• Office BP >130/80 mmHg • Home BP <130/80 mmHg • Management: Lifestyle modifications	• Office BP 120-129/<80 mmHg • Home BP >130/80 mmHg • Management: Start treatment

- Home BPM:
 - Monitor for 7 days, twice daily
 - Discard Day 1 readings, use the average of the next 6 days

CAUSES OF HYPERTENSION

00:20:51

- M/c cause of HTN: Essential/Primary/Idiopathic hypertension
- M/c cause of secondary HTN: **Renal parenchymal disorder** (works more than usual)
- M/c congenital cause: Coarctation of the aorta

UNEQUAL BP BETWEEN RIGHT AND LEFT ARM

Mnemonic: CATS

- **C** - Coarctation of Aorta (Pre-ductal type)
- **A** - Aortic Dissection
- **T** - Takayasu Arteritis
- **S** - Supravalvular Aortic Stenosis

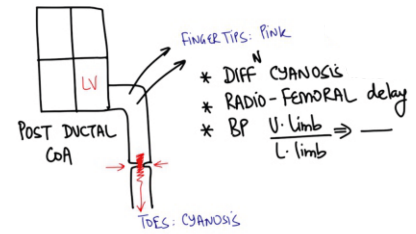
COARCTATION OF AORTA (PRE-DUCTAL TYPE)	• Narrowing of the aorta before the ductus arteriosus → ↓ blood flow to the Left Subclavian artery • Right Subclavian artery (right arm) BP > Left Subclavian artery (left arm) BP • Disparity between the BP values is due to increase in peripheral resistance	PREDUCTAL CoA
TAKAYASU ARTERITIS	• Cause ostial narrowing of Subclavian arteries (LSA > RSA) → Variable perfusion → BP differences between arms	TAKAYASU ARTERITIS

POST-DUCTAL COARCTATION OF AORTA

• Narrowing at the starting of Descending aorta → ↓ in blood supply to legs

• Leads to:

- Differential cyanosis
 - Cyanosis in toes
 - Pink fingertips
- Radio-femoral delay
- Upper limb BP more than Lower limb BP
 - In normal person, on sitting position: LL BP will be higher than UL BP (difference of <20 mmHg) d/t gravity



SECONDARY CAUSES OF HYPERTENSION

00:33:15

- **M/c:** Renal Parenchymal Disease (CKD)
- Renal Vascular Disease:
 - Renal Artery Stenosis (RAS)
 - 20-year-old female → **Fibromuscular dysplasia**
 - 60-year-old male, smoker → **Atherosclerosis**
 - Ostial narrowing / RAS with arm claudication → **Takayasu arteritis**
- Metabolic Syndrome: **Syndromes X**
- Obstructive Sleep Apnea (OSA): Apneic episodes → Hypoxia → Vasoconstriction → Systemic hypertension
- Endocrine Causes:
 - Myxedema ($\downarrow T_4$): Isolated diastolic HTN
 - Thyrotoxicosis: $\uparrow T_4$ → $\uparrow$ sympathomimetic drive → Vasoconstriction
 - Pheochromocytoma
 - Conn's Syndrome (tumour of Zona glomerulosa)
 - Cushing's Syndrome (tumour of Zona fasciculata)
 - Acromegaly
 - Oral Contraceptive Pills → 2° aldosteronism

00:39:42

MENDELIAN CAUSES

→ Liddle Syndrome

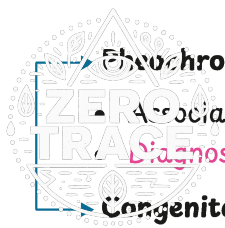
- ENaC (Epithelial Sodium Channel) gain-of-function mutation
- Urinary loss of K^+/H^+
- **Hypertension with hypokalemic alkalosis**
- $\downarrow$ Aldosterone, $\downarrow$ Renin

→ Gordon Syndrome

- Mutations in WNK1, WNK4 → Gain-of-function mutation of Na^+Cl^- co-transporter
- Hypertension: $\downarrow$ Renin, $\downarrow$ Aldosterone
 - D/t $\downarrow$ Aldosterone → $\downarrow$ Potassium excretion → Hyperkalemia

→ Polycystic Kidney Disease

- Autosomal Dominant (AD): USG criteria 30–59 years of age, **>2 cysts/kidney**
- Autosomal Recessive (AR)



→ Pheochromocytoma

- Associated with MEN 2/2A, MEN 3/2B, VHL, NF-1
- **Diagnosis:** 24-hour urinary fractionated metanephrine

→ Congenital Adrenal Hyperplasia

- 17-Hydroxylase deficiency: ↑ Aldosterone
- 11 β -Hydroxylase deficiency: ↑ 11-deoxycortisol → ↑ ENaC virilization activity

WORKUP FOR NEWLY DIAGNOSED HYPERTENSION PATIENTS

00:52:20

- Echocardiography: Check for LVH (Left Ventricular Hypertrophy)
- Fasting Blood Sugar (FBS)
- Lipid profile
- Thyroid Stimulating Hormone (TSH)
- Urine Albumin Excretion Rate (AER):
 - 30–300 mg of albumin per gram of urinary creatinine
- Kidney Function Tests (KFT) and Electrolytes
- Uric acid levels
- ASCVD risk score:
 - Includes age, gender, Diabetes mellitus (DM), smoking status, cholesterol levels, HDL, BP or HTN on medication

TARGET ORGAN DAMAGE

00:52:28

→ Heart

- **First organ involved**
- LVH → ST depression on ECG

→ Eyes

- Hypertensive Retinopathy (Grades):
 - Grade I: Focal attenuation of arterioles (**earliest fundus finding**)
 - Grade II: Grade I + A-V nipping (**Salu sign**)
 - Grade III: Grade I + II + Hemorrhages
 - Grade IV: Grade I + II + III + Papilledema

→ Kidney

- ↓ Glomerular Filtration Rate (GFR)

→ Brain

- Hemorrhagic Stroke- M/c site is the Putamen region

→ Aorta

- Aortic Dissection
 - Chest pain at rest, interscapular area
 - Antegrade dissection: **Horner's syndrome**
 - Retrograde dissection: **Coronary artery dissection**
 - CT Angiography: Tennis ball appearance

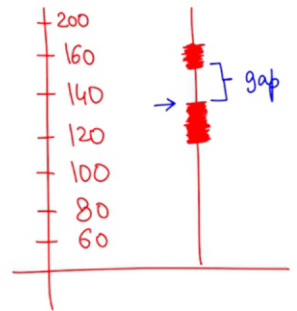
WHEN TO TREAT?

01:02:58

- Indications for starting BP lowering medication:
 - Secondary prevention of recurrent CVD, BP >130/80 mmHg
 - Primary prevention
 - If ASCVD Risk >10%, BP >130/80 mmHg
 - If Low Risk ASCVD, BP >140/90 mmHg
- Target BP <130/80 mmHg (in relatively younger people)

AUSCULTATORY GAP

- Definition: Temporary silence of **Korotkoff** sounds
- Effects:
 - Underestimation of Systolic BP (SBP)
 - Overestimation of Diastolic BP (DBP)
- Occurs due to hardening of blood vessels (in old age due to calcification)



HOW TO AVOID IT?

- First use the palpation method for estimation

HOW TO TREAT?

01:10:41

LIFESTYLE MODIFICATIONS

MODIFICATION	INTERVENTION	RESULTING DECREASE IN BP
Weight reduction	• BMI <25	• 5-20 mmHg/10 kg loss
Salt restriction	• <6 g/day (<100 mmol/day)	• 2-8 mmHg
DASH Diet	• Diet rich in fruits, vegetables, nuts, low-fat dairy	• 8-14 mmHg
Alcohol intake	• Men: ≤2 drinks/day • Women: ≤1 drink/day	• 4 mmHg
Physical activity	• Aerobic 30 minutes/day • Dynamic 90-150 minutes/week • Isometric (hand grip 4 repetitions, 3 times/week)	• 5-10 mmHg
Mindfulness	• Meditation and breathing control	• 5 mmHg

PHARMACOLOGICAL THERAPY

01:13:45

AGE >55 YEARS

- Start with CCB (e.g., Amlodipine)
- If BP is not controlled after 2 weeks: **Add ACE inhibitors**
- If BP is not controlled after 2 weeks of adding ACEi: **Add Thiazide / Spironolactone**
 - K⁺ values are monitored before starting Spironolactone

AGE <55 YEARS

- Start with ACEi or ARB



- S/E of ACE inhibitors and ARB: Hyperkalemia
- If BP is not controlled after 2 weeks: **Add CCB**
- If BP is not controlled after 2 weeks of adding CCBs: **Add Thiazide / Spironolactone**

β BLOCKER INDICATIONS:

- Chronic stable angina
 - Mortality reduction (max) in CSA: Metoprolol
- Chronic heart failure

RESISTANT HYPERTENSION

- Definition: Uncontrolled BP despite >3 classes of anti-hypertensive medication being used concurrently (must include a diuretic)
- Causes:
 - Faulty readings, calibration errors
 - Non-compliance
 - Excess salt intake, alcohol use, obesity
 - COX-1 inhibitors, cocaine, steroids, erythropoietin injection

ANTIHYPERTENSIVE DRUGS

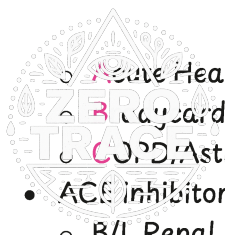
β BLOCKERS (CARDIOSELECTIVE)	• Metoprolol, Bisoprolol, Carvedilol	• HTN with ◦ Chronic Stable Angina ◦ Chronic Heart Failure
ACE INHIBITORS	• ACE inhibitors remain protective and safe in kidney disease associated with significant proteinuria and serum creatine as high as 5 mg/dl	• Post MI • ACS • HF with reduced ejection fraction (HFrEF) • Diabetic Nephropathy
ALDOSTERONE ANTAGONISTS (AA)	• Spironolactone, Eplerenone	• HFpEF, HFrEF
α BLOCKERS	• Prazosin	• HTN with BPH • Pheochromocytoma (Phenoxybenzamine) • ESRD
CCB	• Amlodipine	• HTN with Asthma/COPD

Important Information

- HFrEF treatment = ACE inhibitors + β-blocker + Aldosterone Antagonist
 - Major mortality reducing agent: **ACE inhibitors**
- Major mortality reducing agent post MI: **ACE inhibitors**

CONTRAINDICATIONS

- β Blockers are contraindicated in:



• Acute Heart Failure

• Brugada (High-grade heart block)

• COPD/Asthma

- ACE inhibitors are contraindicated in:
 - B/L Renal Artery Stenosis
 - Pregnancy (Risk of Renal agenesis)
- Do not combine
 - Verapamil and Metoprolol → ↓ HR
→ Verapamil causes prolongation AV nodal delay
 - ACE inhibitors and ARB → ↑ K⁺

PREFERRED DRUGS FOR FOLLOWING HYPERTENSIVE EMERGENCIES

- Hypertensive Emergency: BP >180/120 mmHg with target organ damage
- Hypertensive Urgency: BP >180/120 mmHg, no target organ damage
 - Managed by gradual lowering of BP over 24 hours by oral ACEi

HYPERTENSIVE EMERGENCY	DRUGS
Hypertensive encephalopathy	Nitroprusside
Malignant hypertension	Labetalol, Nicardipine
Stroke	Nicardipine
Adrenergic crisis	Phentolamine, Nitroprusside
Acute LVF/Post-op HTN/MI/Unstable angina	Nitroglycerin
Aortic dissection	Nitroprusside, Esmolol

- Goal BP (Emergency): up to 25% lowering within minutes to 2 hours (target ~160/110 mmHg)



Q. Which of the following conditions causes Hypertension with Hyperkalemia?

- a. Liddle syndrome
- b. Gordon syndrome
- c. Conn Syndrome
- d. Cushing Syndrome

Ans: (b)

Q. Cut off for grade II Hypertension as per AHA guidelines

- a. >130/80 mmHg
- b. >140/90 mmHg
- c. >180/120 mmHg
- d. >220/130 mmHg

Ans: (b)

Q. Which of the following drugs is best to control BP of 160/100 mmHg in patient of ESRD with pedal edema. Work up shows Serum K^+ : 6 mEq/L, Na^+ 130 mEq/L and uric acid 10 mg/dl.

- a. Amlodipine
- b. Hydrochlorothiazide
- c. Prazosin
- d. Ramipril

Ans: (c)



DISEASES OF PERICARDIUM

DISEASES OF PERICARDIUM

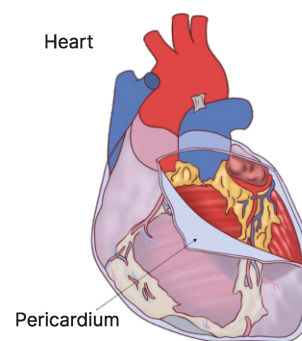
ACUTE PERICARDITIS

- Symptoms develop < 6 weeks

CAUSES

00:01:50

- Idiopathic
- Infections
 - Tuberculosis (Extrapulmonary) - Tubercular pericarditis
 - Viral pericarditis - Coxsackie A and B, Echovirus, Herpes, HIV
- Collagen vascular diseases
 - SLE
 - M/c cardiovascular involvement in SLE - Pericarditis
 - Rheumatoid arthritis
 - Rheumatoid nodules → chest pain
- Autoimmune pericarditis - Post MI (3-6 weeks)
 - Dressler syndrome
 - T/t - Aspirin
- Malignancies
 - Oat cell cancer (lung), Breast cancer
- Drugs
 - Isoniazid, Procainamide
- Renal failure
 - ↓ eGFR = 15 ml/ min- End Stage Renal Disease
 - Complication - Uremic pericarditis
- Radiation to chest
 - Indication - Hodgkin's lymphoma, recurrence of breast cancer
 - Radiation administered → damages outermost layer of heart → pericarditis

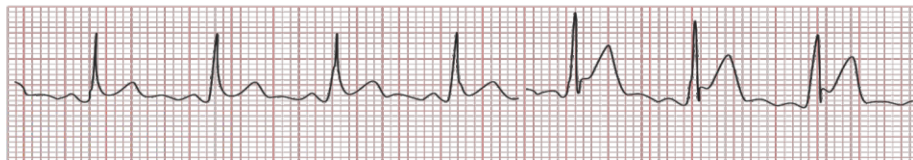
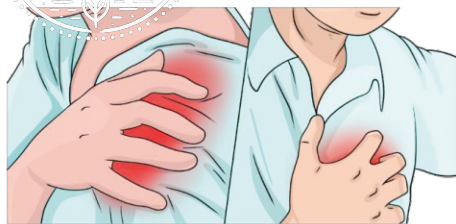


Important Information

- Rheumatic Pericarditis
 - Pediatric presentation - Child with chest pain
 - Inflammation - Due to Aschoff nodules

CASE SCENARIO

Young man presents with diffuse central chest pain with radiation to left shoulder and relief on sitting and leaning forward

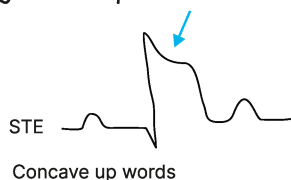


ON AUSCULTATION

- Pericardial friction rub -
 - Patient is asked to hold breath for 5 seconds (Deep inspiration/expiration breath holding)
 - Pericardial friction rub persists
 - Pleural rub would disappear
- S1, S2 - Normal
- Clear B/L lung fields

INVESTIGATIONS

- ECG -
 - ST elevation (concave upwards)
 - PR segment depression



- ECHO - Effusions (of variable size) +/-
- Biomarkers - Mild elevation
 - Possibility of myopericarditis

DIAGNOSIS

- Acute pericarditis

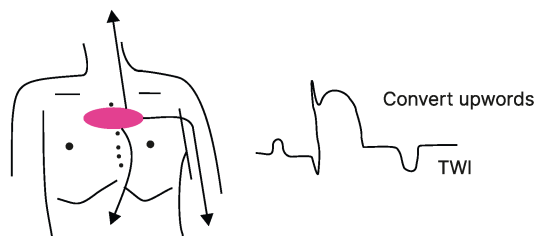
Important Information

- Nerve responsible for radiation of pain to left shoulder (pericarditis) - Phrenic nerve

STEMI

00:20:03

- Levine sign
- Radiation of pain towards
 - Left arm, angle of jaw, Epigastrium
- ST-segment elevation
 - Convex upwards, T wave inversion
- Cardiac Troponin I level -
 - > 99th percentile of Upper Reference Limit



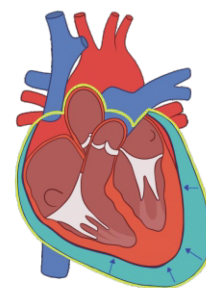


TYPE	T/t
Viral pericarditis	NSAIDs
TB pericarditis	ATT
Dressler Syndrome	Aspirin
Uremic pericarditis	Hemodialysis (absolute indication)
Malignancy (Oat cell cancer)	Chemotherapy

PERICARDIAL EFFUSION

00:23:16

- Normal pericardial fluid - 20-50 ml
 - Pericardial effusion - > 50ml fluid
- ECG findings - **Low voltage** ECG
 - Height of R + depth of S
 - < 5 mm in Limb leads
 - < 10 mm in Chest leads
 - Electrical alternans
- IOC - Echocardiography



Important Information

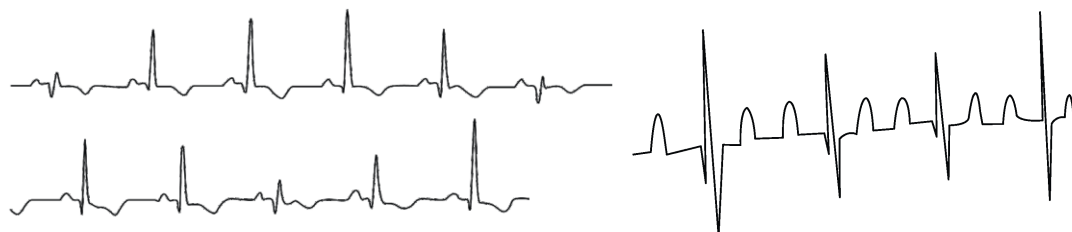
- Causes of low voltage ECG -
 - Myxedema
 - Restrictive cardiomyopathy (Fibrosis)
 - Constrictive pericarditis (Calcification)

ELECTRICAL ALTERNANS

Seen in
pericardial effusion

- Fluctuation in amplitude of QRS complex
 - Cause - **Swirling motion** of heart (blood pumping in pericardial effusion)
 - Beat-to-beat variation

QRS duration, PR interval, ST segment -
Normal





POLYMORPHIC VENTRICULAR TACHYCARDIA

Variation of QRS amplitude

Broad QRS complex > 120 msec

Development of prolonged QT interval predisposes to polymorphic ventricular tachycardia.



CHEST X RAY

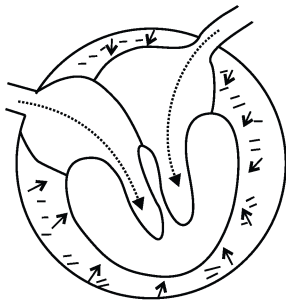
FINDINGS

↑ Cardiothoracic ratio
CARDIAC TAMPONADE

Narrow vascular pedicle

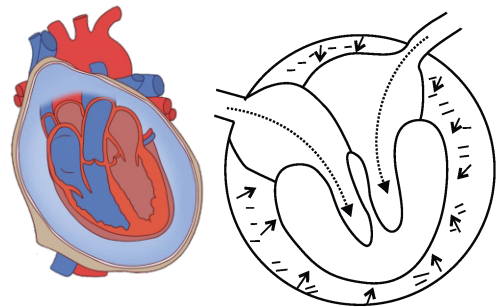
Water bottle heart/ Money bag appearance
00:34:45

LOW PRESSURE CARDIAC TAMPONADE

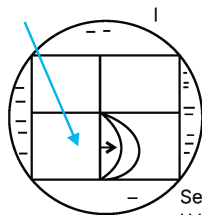


- Atrial collapse - ↑ intrapericardial pressure crush atria
- 30% ↓ filling of ventricles → 30% ↓ in End Diastolic Volume → 30% ↓ in Stroke Volume → 30% ↓ in Systolic BP
 - ↓ Pulse pressure - Feeble pulse
- Progress to cardiac tamponade
- Pulsus paradoxus **not seen**

CARDIAC TAMPONADE

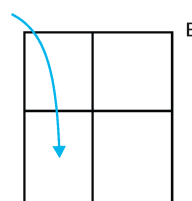


- Atrial + Ventricular collapse
- ↓↓ End Diastolic Volume
- ↓↓ Stroke Volume
- ↓↓ Systolic BP
- Obstructive shock
- Pulse disappears due to ↓↓ filling
- **Pulsus paradoxus seen**
 - Heart is beating but pulse disappears (inspiration phase)
- ECG - Electrical alternans



Reverse Berheim Sign

Septal deviation ++
LV cavity size ↓↓
EDV ↓↓ SV ↓↓
SBP: 115mm Hg
105



LV Cavity n
EDV n SV n
SBP: 120mm Hg



CHANGES

NORMAL INDIVIDUAL

- Blood entering (into right ventricle) is higher during inspiration (compared to expiration)
 - Cause - Negative pressure arising on inspiration
- Leads to mild septal deviation → ↓ LV cavity size, ↓ End Diastolic Volume, ↓ stroke volume
 - ↓ SBP = 115mm Hg
- Relatively less blood enters during expiration → septal deviation disappears → normalization of LV cavity size, EDV, SV and SBP (120 mm Hg)

CARDIAC TAMPONADE

- Exaggerated septal deviation present → ↓↓ in LV cavity size, ↓↓ End Diastolic Volume, ↓↓ stroke volume
 - Systolic BP - 105mm Hg
- Normal inspiratory expiratory variation of systolic blood pressure <10 mm Hg
- If inspiratory expiratory variation of systolic blood pressure ≥ 12 mm Hg
 - Pulsus paradoxus
- Reverse Berheim sign

CAUSES OF CARDIAC TAMPONADE

Trauma Aortic dissection Pericarditis - Oat cell lung carcinoma MI (free wall rupture) TB

- Examination of Pulsus Paradoxus - Done in normal breathing

CAUSES OF PULSUS PARADOXUS

Cardiac tamponade Constrictive pericarditis Status asthmaticus Acute cor pulmonale/ Pulmonary embolism Pregnancy Hypovolemic shock

T/t OF CARDIAC TAMPONADE

00:53:00

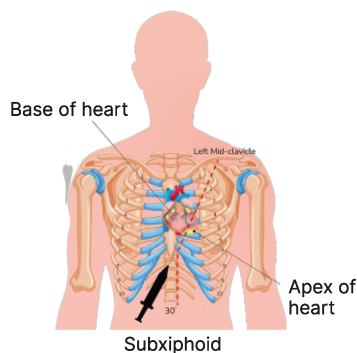
T/t

TOC

Due to penetrating trauma

ECHO-guided pericardiocentesis
-Xiphisternal approach

Emergency resuscitative thoracotomy



Emergency resuscitative
Thoracotomy

MASSIVE PERICARDIAL EFFUSION

00:55:14

- Seen in hypothyroidism - Myxedema heart
- Lymphedema
- Gradual accumulation of fluid in pericardial space
- Heart gradually adjusts to this gradual rise in pressure → **Absent** Pulsus paradoxus

Important Information

- Pulsus paradoxus is not seen in
 - Massive pericardial effusion
 - Low-pressure cardiac tamponade

CASE BASED SCENARIO

A 25-year male driving a car under the influence of alcohol was not wearing a seatbelt and has a car accident. Ambulance crew informs you on phone that the patient is breathing, pulse rate: 130/min, BP: 80/60 mmHg, cervical collar is in place and IV cannula is put and fluid bolus is given. Patient is declared dead on arrival. What is the cause of death?

- a. Hemopericardium
- b. Hemothorax
- c. Hemoperitoneum
- d. Vertebral artery dissection

Ans: **(a)**

- Vertebral artery dissection is ruled out
 - Causes brain stem infarction → Cessation of breathing
- Hemoperitoneum and hemothorax is ruled out
 - Both will cause hypovolemic shock → will always improve on IV bolus given
- Hemopericardium
 - Causes obstructive shock
 - Initial step for management - Pericardiocentesis
- Seatbelt - → Hemopericardium
- Seatbelt + → **Seatbelt syndrome**
 - Mesenteric tear - Damage to branches of superior & inferior mesenteric arteries

CONSTRUCTIVE PERICARDITIS

01:02:57

- Sequelae of -
 - TB Pericarditis, Pyopericardium
- ↓ Compliance Of Ventricles - due to calcification

RIGHT VENTRICLE COMPLIANCE ↓	LEFT VENTRICLE COMPLIANCE ↓
• Pedal edema • Ascites • Hepatomegaly/ Right Upper Quadrant discomfort	• Pulmonary edema ○ Shortness of breath, Orthopnea • Effort intolerance

ON EXAMINATION

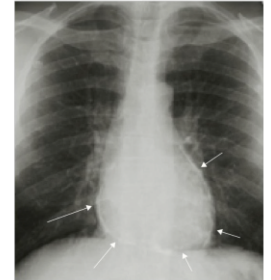
• Pericardial shudder/ knock/ shock

• Kussmaul sign

- Paradoxical rise of JVP on inspiration
- Seen in constrictive pericarditis, restrictive cardiomyopathy, right sided CHF
- Steep 'x', Steep 'y' descent
- S4 +
- Pulsus paradoxus

WORK UP

- Chest X-Ray - Normal Cardio-thoracic ratio, Egg in cup appearance
- ECG- Low voltage ECG
- ECHO / Doppler / Cardiac Catheterization - **Square root wave** sign
- Cardiac MRI (**IOC**) / CT Chest - Calcification outside heart
- Endomyocardial Biopsy - differentiate restrictive cardiomyopathy from constrictive pericarditis



D/D

- Restrictive cardiomyopathy
 - Stiffened ventricle
 - Pericardium not affected



MRI

T/t

- Pericardial stripping procedure

Important Information

- Kussmaul sign - not seen in cardiac tamponade
 - JVP becomes non-pulsatile
- **Beck's triad** - Muffled S1, S2, Hypotension, JVP - ↑, non-pulsatile
 - Seen in cardiac tamponade



Q. A 26-year-old male patient, an accident victim, started developing difficulty in breathing. On examination, his BP as 80/40 mmHg, and pulses were feeble, JVP was found to be elevated, and heart sounds were distant. What is the most likely diagnosis?

- a. Constrictive pericarditis
- b. Cardiac tamponade
- c. Acute pericarditis
- d. Cardiac failure

Ans: (b)

Q. A patient has presented with features of cardiac tamponade. Which of the following is correct regarding this condition?

- a. Pulsus paradoxus is always present
- b. Kussmaul sign is always present
- c. Tall A wave is present
- d. Jugular veins column is visible distended without pulsations

Ans: (d)

Q. A 65-year-old female patient presents to the OPD with complaints of shortness of breath and tiredness. On examination, there were engorged neck veins and JVP showed steep x and steep y descent. Which of the following is the most likely diagnosis?

- a. Constrictive pericarditis
- b. Cardiac tamponade
- c. Dilated cardiomyopathy
- d. Ischemic cardiomyopathy

Ans: (a)



CONGESTIVE HEART FAILURE

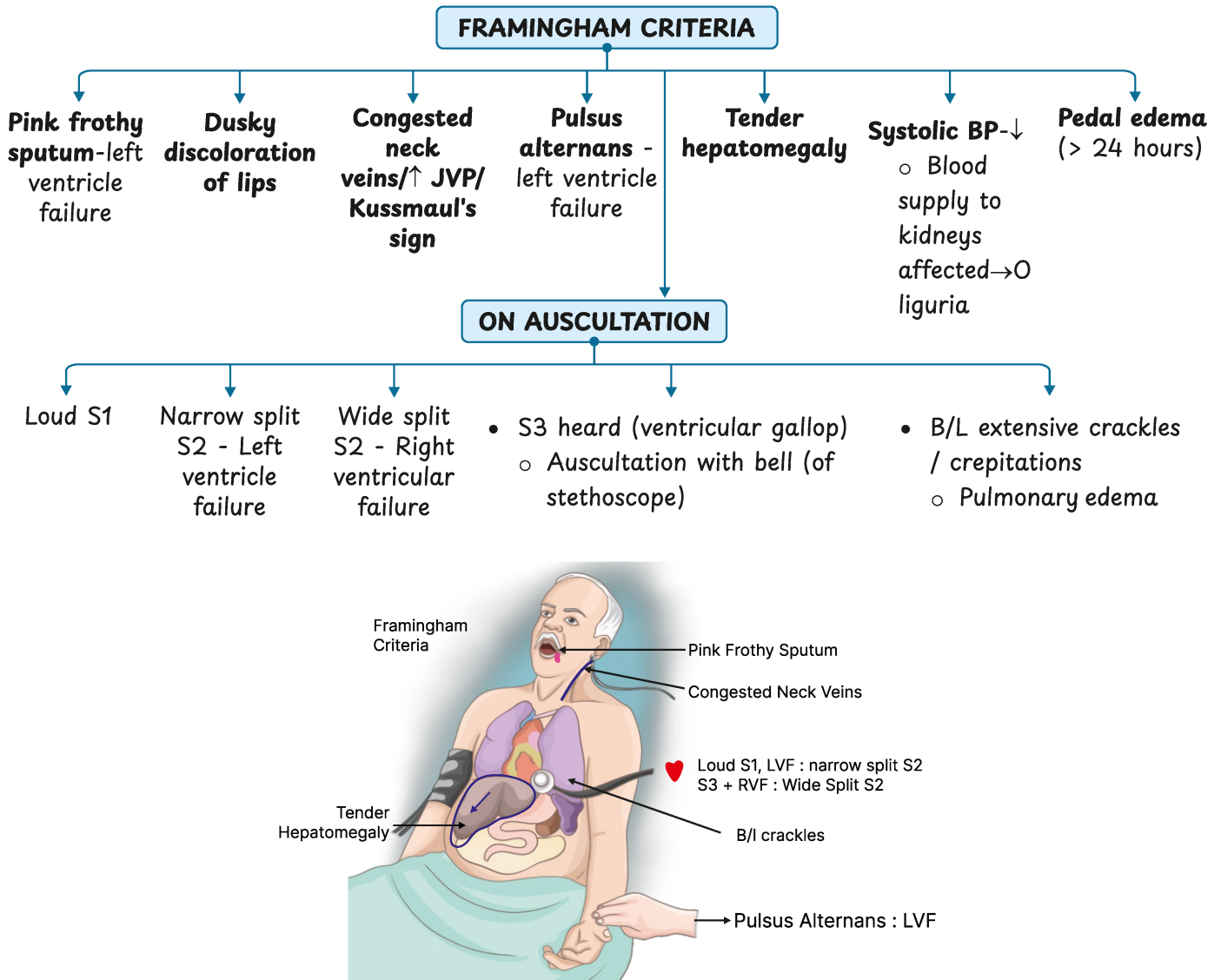


CONGESTIVE HEART FAILURE

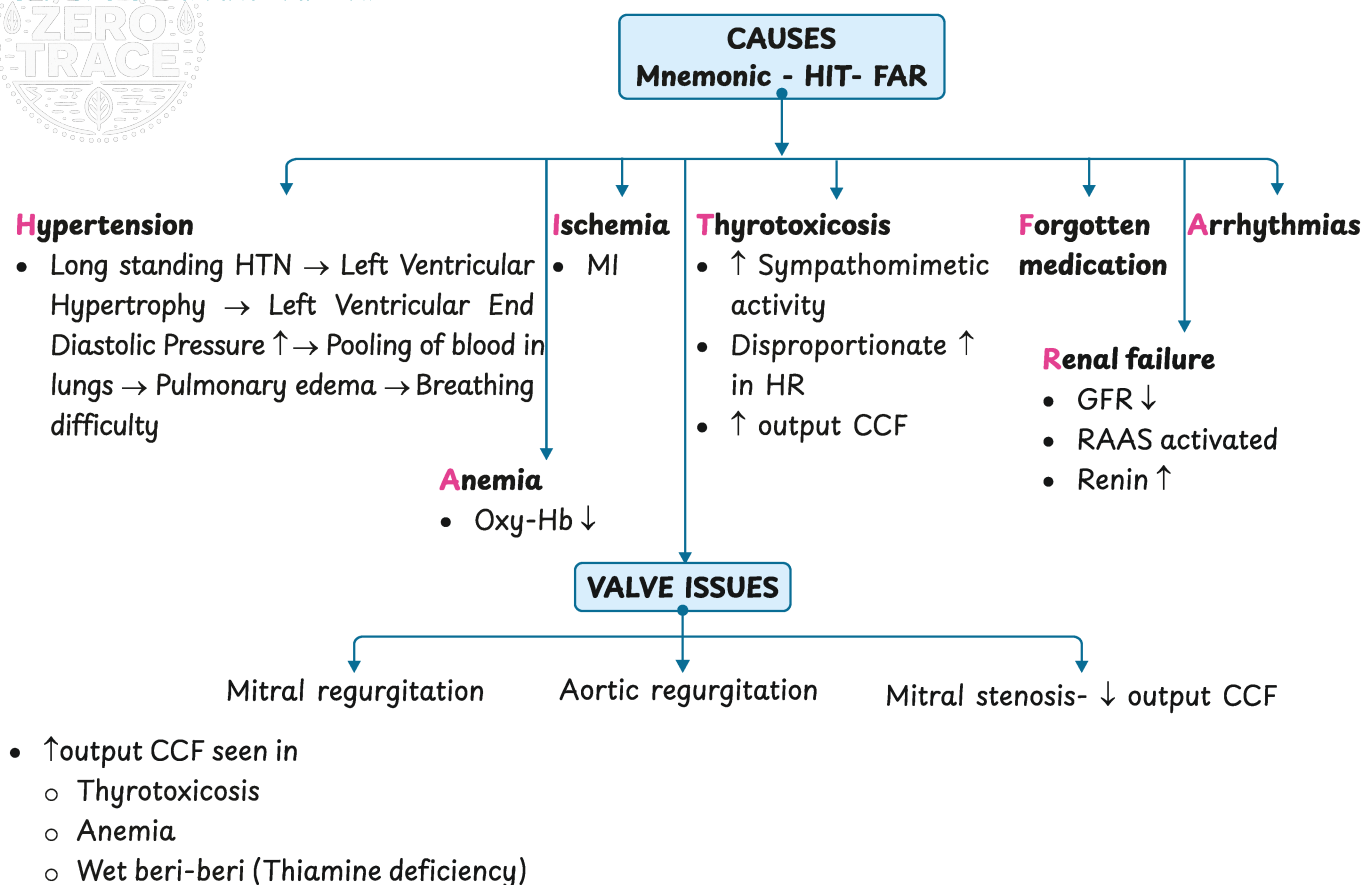
CLINICAL FEATURES

00:00:35

- Acute decompensated heart failure - To be identified



- **Cheyne Stokes breathing**
 - Not part of diagnostic criteria
 - Associated finding
 - Seen in
 - $\rightarrow \uparrow$ ICP, $\uparrow$ voltage pulmonary edema, CHF



MECHANISM

- **Cardiac output (CO)** = Heart rate (HR) × Stroke volume (SV)
 - Changes -
 - ↓ SV, ↑ HR, ↑ sympathetic drive

SYMPTOMS

- **Palpitations**
- **Diaphoresis**
- **Shortness Of Breath / Dyspnea on minimal exertion/ Paroxysmal nocturnal dyspnea**
 - Orthopnea - Dyspnea ↑ in supine position
 - Cause - Pulmonary edema
- **Nocturnal cough**
- **Disturbed sleep**
 - Risk factor - Obesity
- **Effort intolerance**
 - Excessive fatigue
 - Cause - forward failure
- **Nocturia**
 - Patient moving around → peripheral pooling of blood
 - In supine position (night time) → peripheral blood moves to intravascular compartment → perfusion of heart improves → perfusion of kidney improves → ↑ urine production